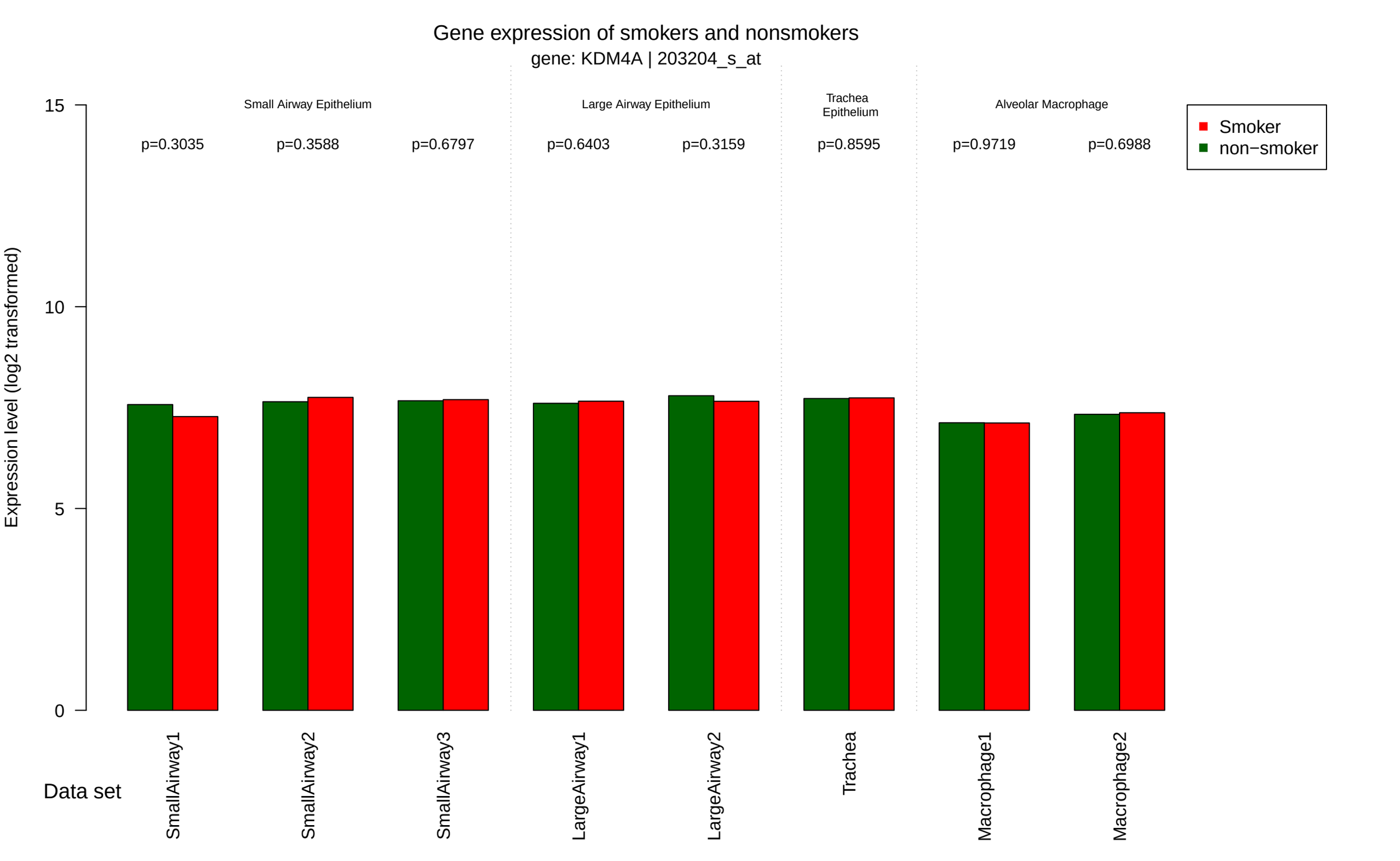
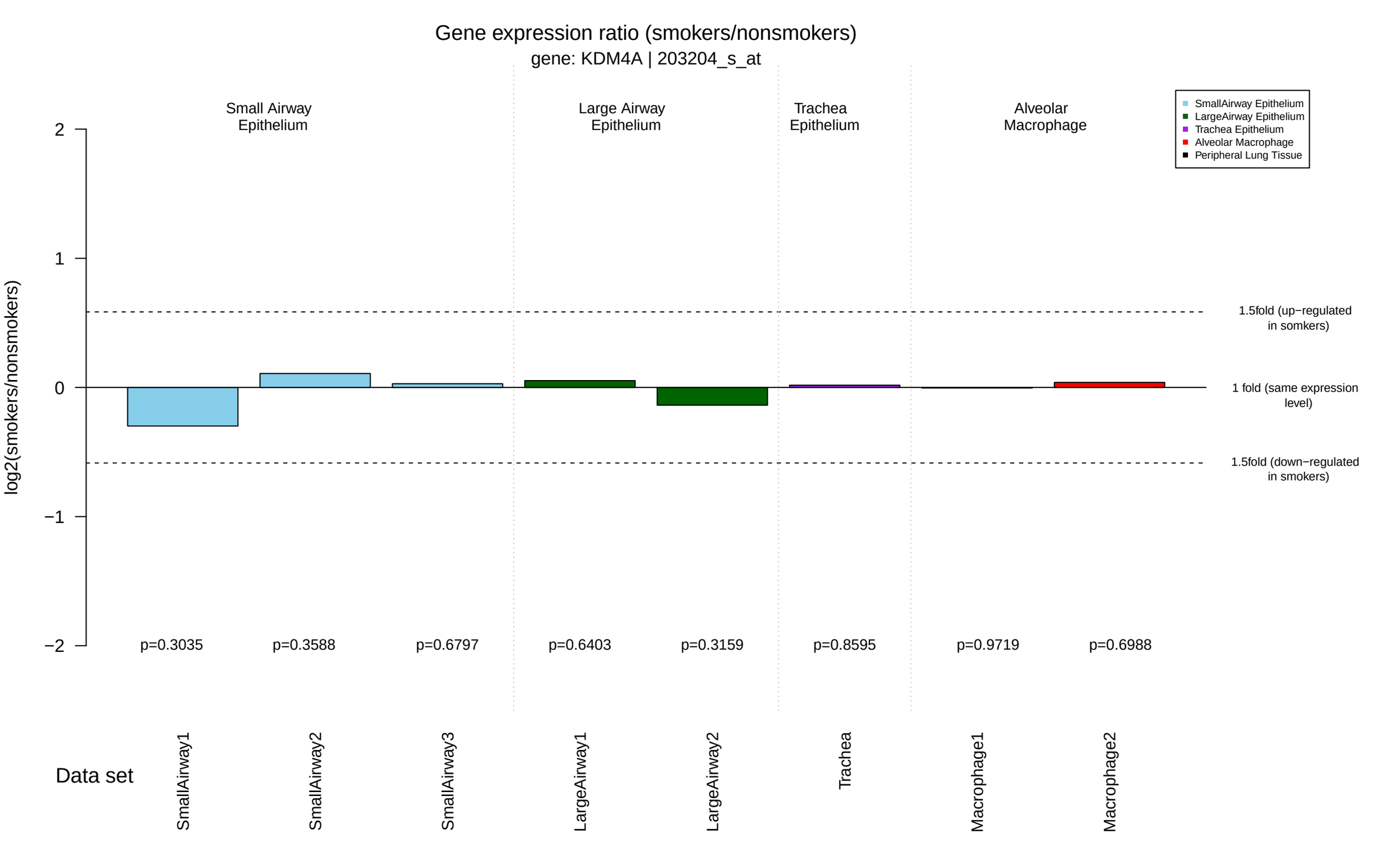
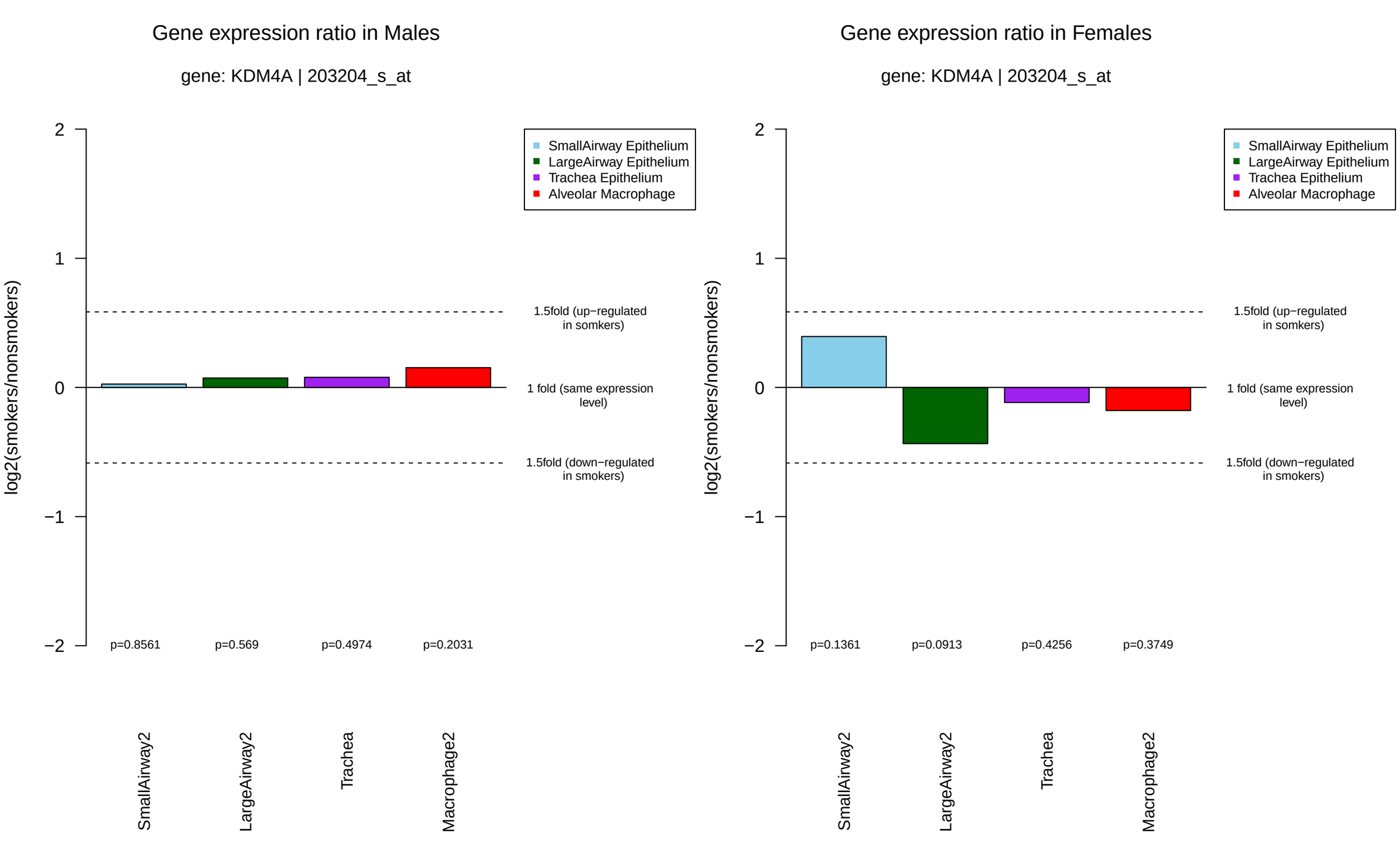
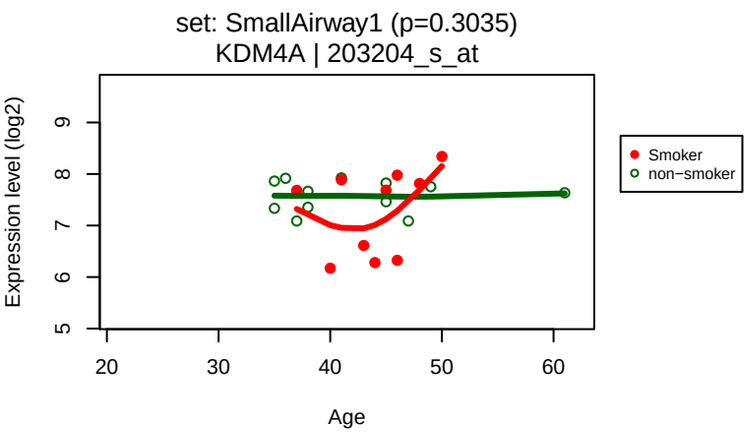
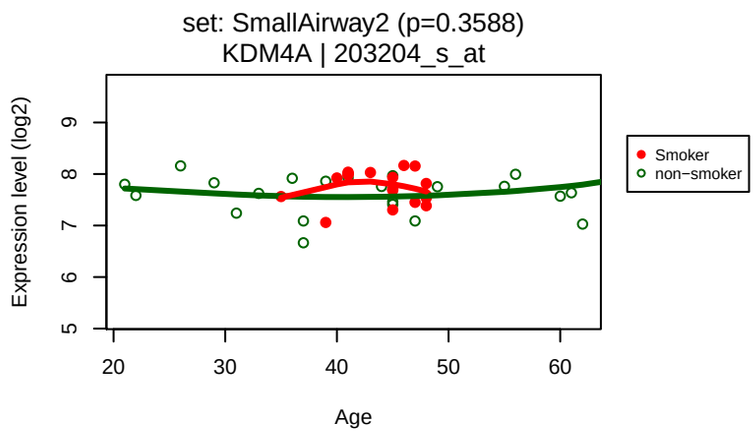
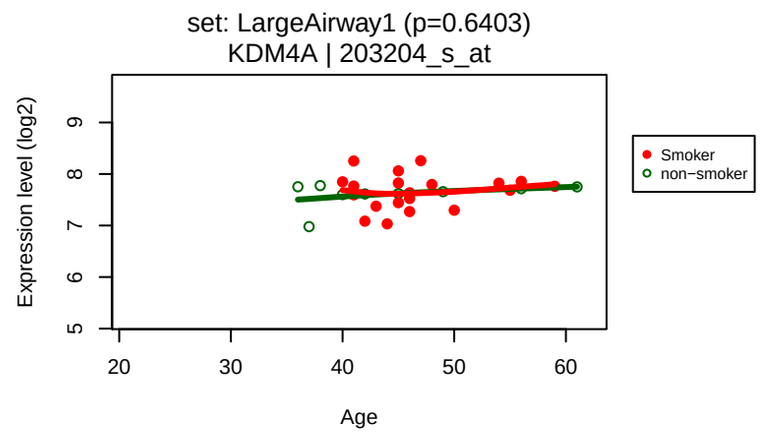

Gene expression of smokers and nonsmokers
gene: KDM4A | 203204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3035
p=0.3588
p=0.6797
p=0.6403
p=0.3159
p=0.8595
p=0.9719
p=0.6988
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

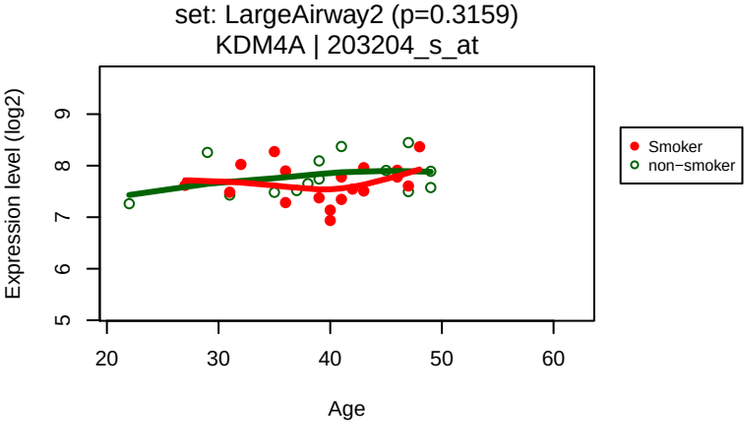

Gene expression ratio (smokers/nonsmokers)
gene: KDM4A | 203204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3035
p=0.3588
p=0.6797
p=0.6403
p=0.3159
p=0.8595
p=0.9719
p=0.6988
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

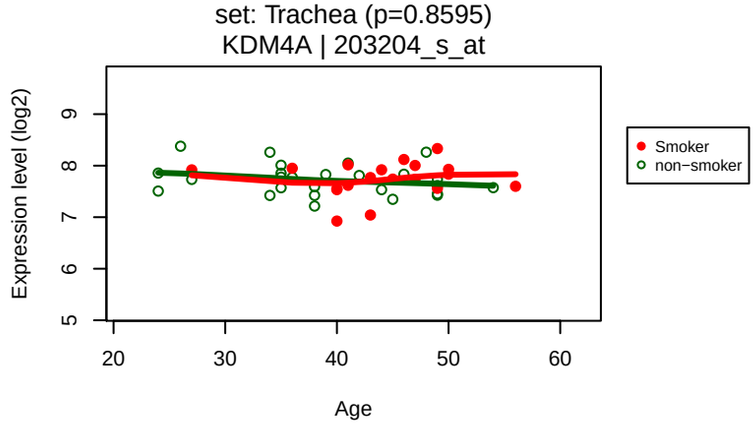

Gene expression ratio in Males
gene: KDM4A | 203204_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8561
p=0.569
p=0.4974
p=0.2031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KDM4A | 203204_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1361
p=0.0913
p=0.4256
p=0.3749
SmallAirway2
LargeAirway2
Trachea
Macrophage2

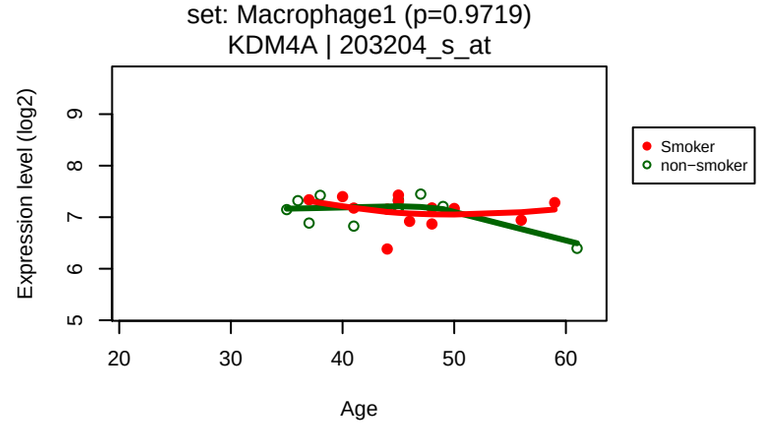

set: SmallAirway1 (p=0.3035)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker

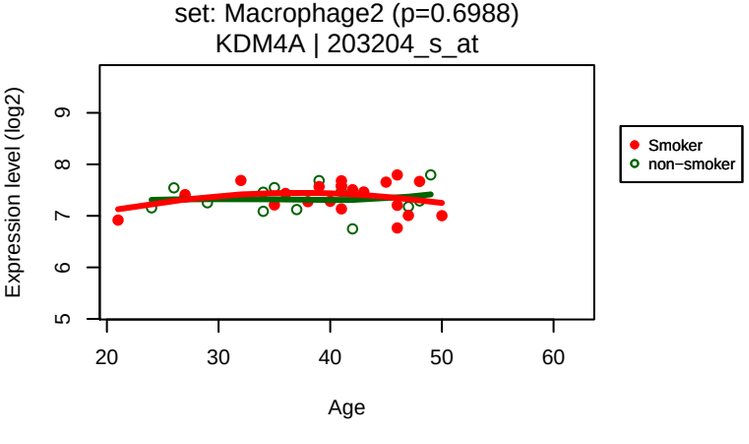

set: SmallAirway2 (p=0.3588)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker

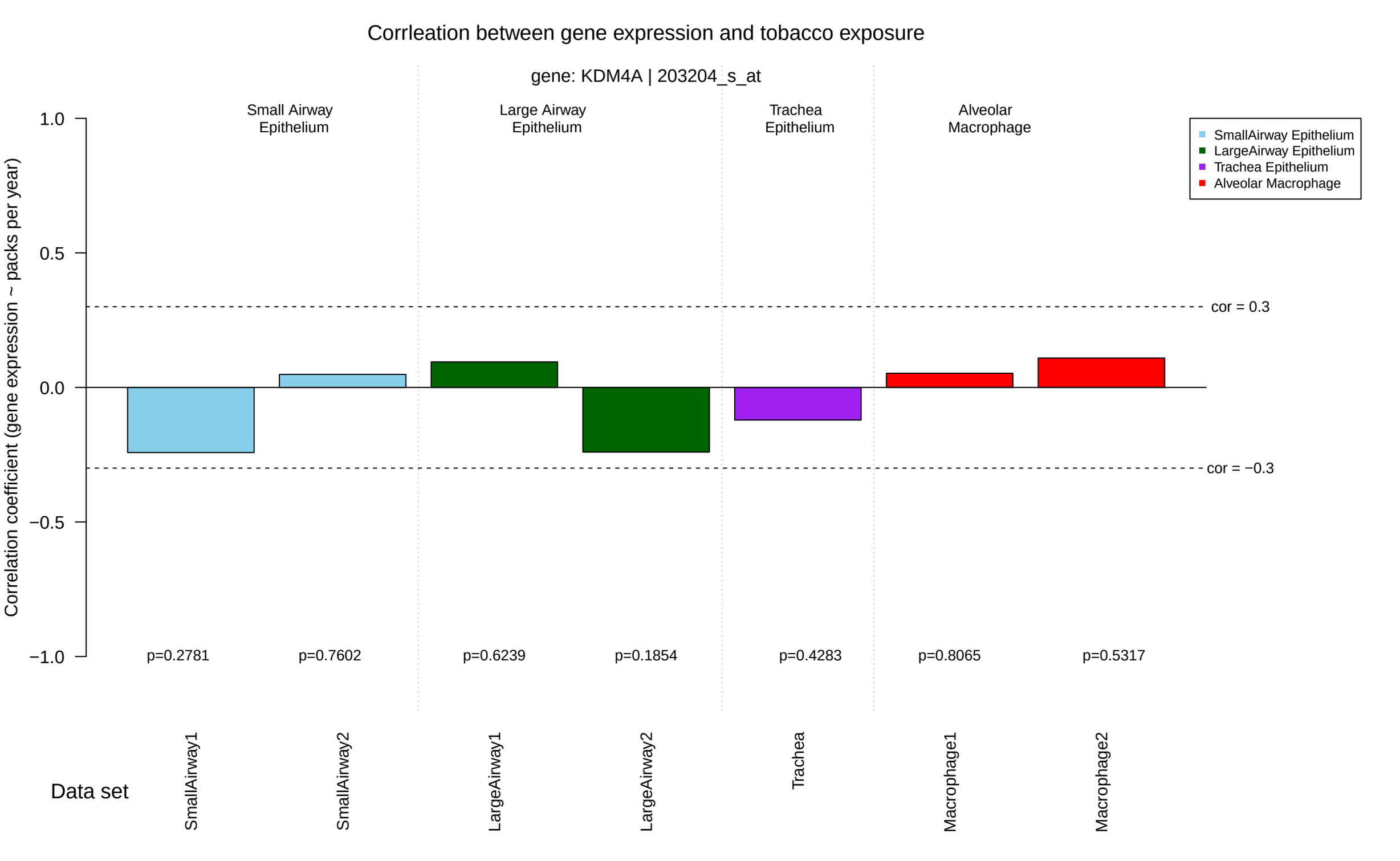

set: LargeAirway1 (p=0.6403)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3159)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8595)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9719)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6988)
KDM4A | 203204_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KDM4A | 203204_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2781
p=0.7602
p=0.6239
p=0.1854
p=0.4283
p=0.8065
p=0.5317
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2